UES301: ANALOG CIRCUITS

L	T	P	Cr
2	0	2	3.0

Course Objectives: To analyse working of BJT biasing circuits and MOSFET, understand multi-stage and power amplifiers, Analysis of operational amplifiers, active filters and oscillators.

Syllabus

Biasing and Thermal Stabilization: Load line analysis, Transistor biasing, Thermal Runaway, Bias Stabilization, Structure, working, output and transfer characteristics of JFET and MOSFET.

Amplifiers: RC coupled Amplifier: Working, Analysis (Calculation of Voltage Gain, Current Gain, Input and Output Impedance), Frequency Response, Power Amplifiers: Class A, Class B, Class AB, and Class C, Calculation of Maximum Efficiency, Power Dissipation, Push-Pull configuration and elimination of Crossover Distortion.

Operational Amplifiers: Introduction and characteristics of op amp, Inverting/Non-inverting amplifiers, Summing/Difference amplifiers, Integrators, and Differentiators, Comparators, Schmitt Trigger (Regenerative Comparator), and Peak Detectors, Half-wave and Full-wave Precision Rectifiers.

Active filters and Oscillators: Design of active filters using op amp, condition for sustained oscillation, R-C phase shift, Hartley, Colpitts, Crystal and Wien Bridge Oscillators.

Specialized ICs & Data Converters: 555 Timer: Internal block diagram, Monostable and Astable multivibrator operations, Voltage Regulators: Fixed (78xx/79xx series) and Adjustable (LM317) regulators.

Laboratory Work

RC coupled amplifier in CE mode, Operational amplifiers Circuits, Butterworth filters and Oscillator circuits,

Minor Project: Automatic intensity control of street lights, auto night lamp with high power LED, water level alarm, panic alarm, dc motor control using L298N motor driver module and Arduino.

Course Learning Outcomes (CLOs) /Course Outcomes (COs):

On completion of this course, the students will be able to:

1. Design different types of transistor biasing circuits
2. Analyse RC coupled Amplifier and compare Power Amplifiers
3. Design and analyze operational amplifier circuits
4. Design Butterworth active filters and oscillator circuits
5. Comprehend the operation of specialized ICs and analyze data conversion techniques

Text Books:

1. Boylestad R. L., Electronic Devices and Circuit Theory, Pearson Education(2007).
2. Millman, J. and Halkias, C.C., Integrated Electronics, Tata McGraw Hill(2006).

Reference Books:

1. Neamen, Donald A., Electronic Circuit Analysis and Design, McGraw Hill(2006).
2. Sedra A. S. and Smith K. C., Microelectronic Circuits, Oxford University Press(2006).

Evaluation Scheme

Evaluation Elements	Weightage %
Mid Semester Test (MST) and End Semester Examination (ESE)	**70**
Sessional may include assignments, regular lab assessments, quizzes, and minor projects (report, presentation etc.)	**30**